

VERILOG מעבדה 1-

מגישים

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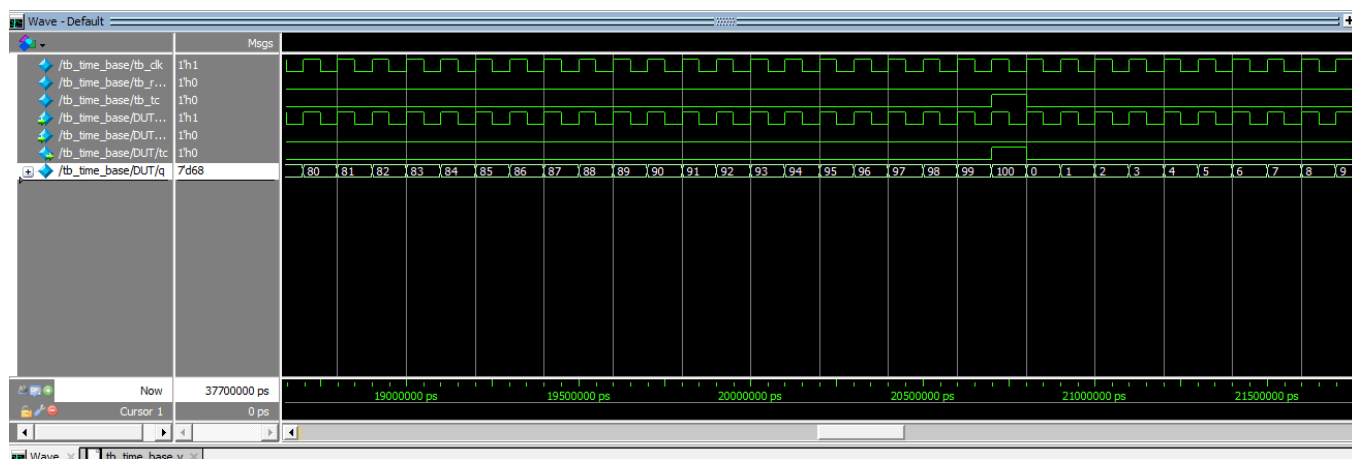
התוכנית:

TB

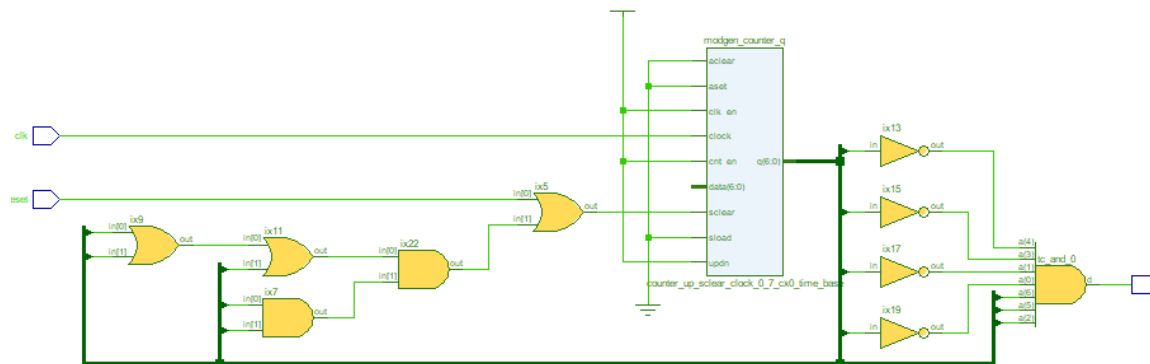
```
1 `timescale 1ns/1ps
2
3 module tb_time_base
4 (
5 );
6
7 reg tb_clk, tb_reset;
8 wire tb_tc;
9
10 initial begin
11     $display ("time\t\t clk reset q tc");
12     $monitor ("%8g\t\t %b %b %b %b",
13         $time, tb_clk, tb , tb_tc);
14
15     tb_clk      = 1;
16     tb_reset    = 1;
17     #700 tb_reset = 0;
18     #5000 $finish;
19 end
20
21 always begin
22     #50 tb_clk = ~tb_clk;
23 end
24
25 time_base DUT
26 (
27     .clk      (tb_clk),
28     .reset    (tb_reset),
29     .tc       (tb_tc)
30 );
31
32 endmodule
```

```
1 module time_base
2 (
3     input wire clk,
4     input wire reset,
5     input wire tc
6 );
7
8 reg [6:0] q;
9
10 always @( posedge clk)
11     if(reset)
12         q <= 7'b0 ;
13     else if (q < 7'd100)
14         q <= q + 1 ;
15     else
16         q <= 7'b0;
17
18 assign tc = (q == 7'b100) ? 1'b1 : 1'b0 ;
19
20 endmodule
```

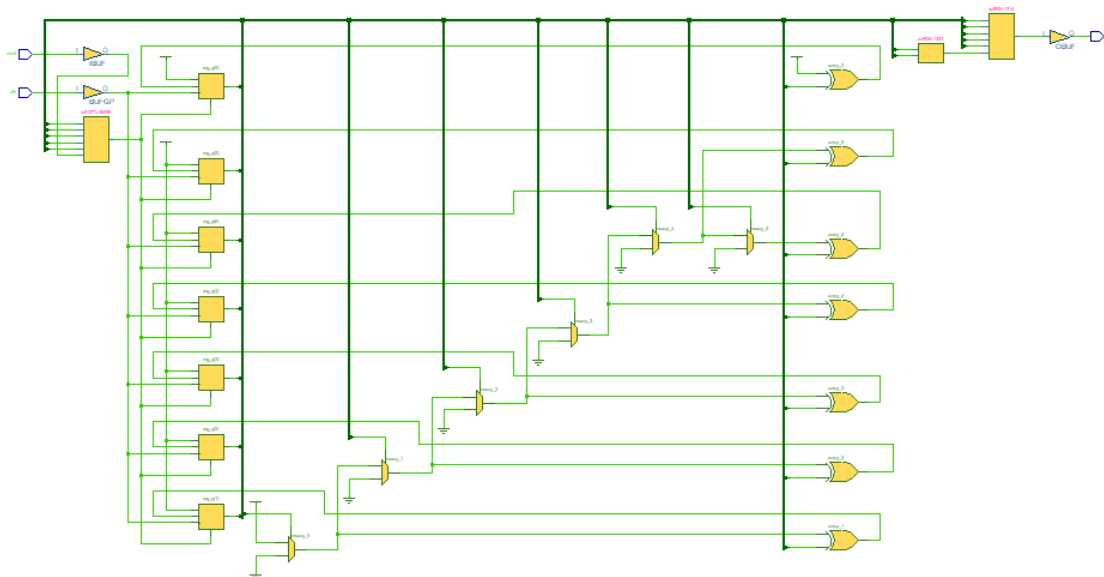
הסימולציה:



RTL- precision:



Technology chematics- precision:



xdc:

```
#####
# Mentor Graphics Corporation
#
#####

#####
# create clock #
#####

# Precision Generated
create_clock [get_ports {clk}] -name {clk} -period 10 -waveform {0 5}
create_clock -name {VirtualClock} -period 10 -waveform {0 5}

#####
# set input delay #
#####

# Precision Generated
set_input_delay 0 -clock [get_clocks {VirtualClock}] -add_delay [get_ports {reset}]

#####
# set output delay #
#####

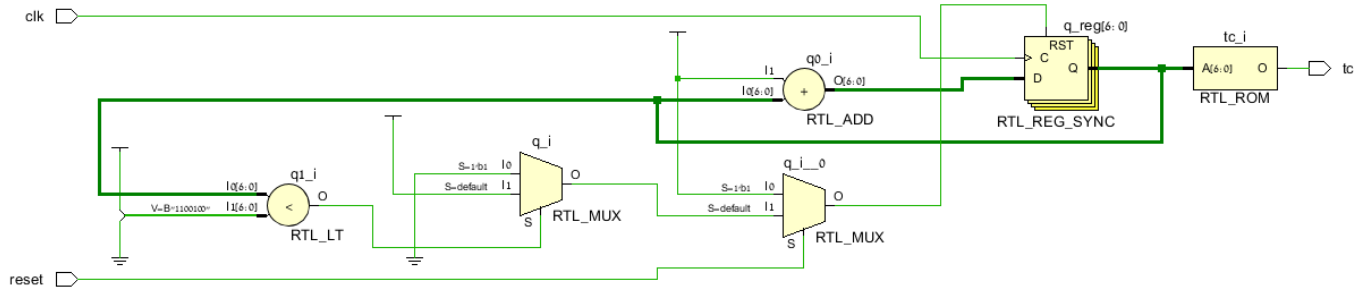
# Precision Generated
set_output_delay 0 -clock [get_clocks {VirtualClock}] -add_delay [get_ports {tc}]

#####
# Pin Locations and Voltages #
#####
set_property PACKAGE_PIN W5 [get_ports {CLK}]
set_property PACKAGE_PIN U18 [get_ports {RESET}]
set_property PACKAGE_PIN H1 [get_ports {TC}]

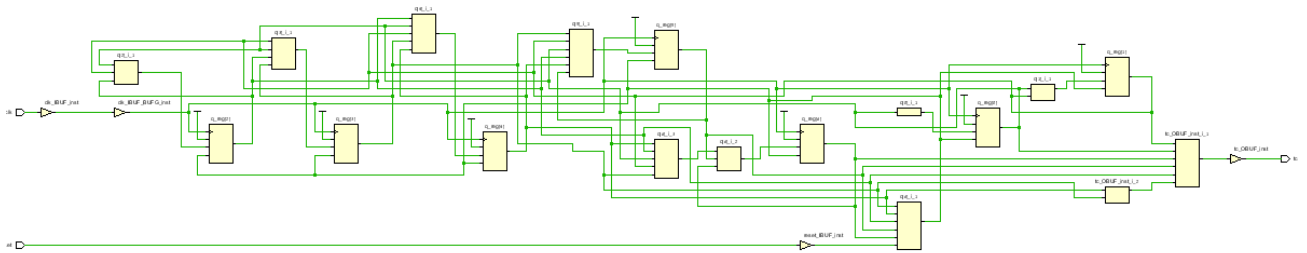
set_property IOSTANDARD LVCMOS33 [get_ports { CLK}]
set_property IOSTANDARD LVCMOS33 [get_ports {RESET}]
set_property IOSTANDARD LVCMOS33 [get_ports {TC}]

#####
# Configuration Settings #
#####
set_property CFGBVS VCCO [current_design]
set_property CONFIG_VOLTAGE 3.3 [current_design]
```

RTL VIVADO:



Technology chematics- VIVADO



Device View:

